

QFT1 Formulas

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1 A scalar with a higher-derivative kinetic term

1.1 The scaling dimension of the coupling constant

Since the action is a scalar, and must have $[S] = 0$, then the Lagrangian must have $[\mathcal{L}] = d$. The kinetic term then has:

$$\left[\frac{1}{2} ((\partial_\mu \partial^\mu + m^2)\phi)((\partial_\nu \partial^\nu + m^2)\phi) \right] = 2 \cdot 2 + 2[\phi] = d \quad (1)$$

$$[\phi] = \frac{d}{2} - 2 \quad (2)$$

The interaction term, then, has a dimension:

$$\left[\frac{\lambda}{4!} \phi^4 \right] = [\lambda] + 4[\phi] = d \quad (3)$$

$$[\lambda] + 4 \left(\frac{d}{2} - 2 \right) = d \quad (4)$$

$$[\lambda] = 8 - d \quad (5)$$

Sanity check:

$d = 4$	$[\phi] = 0$	$[\lambda] = 4$
$d = 6$	$[\phi] = 1$	$[\lambda] = 2$
$d = 8$	$[\phi] = 2$	$[\lambda] = 0$

1.2 Additional terms for the $m = 0$ case

For $m = 0$, the theory reduces to

$$\mathcal{L} = \frac{1}{2} (\partial^\mu \partial_\mu \phi) (\partial^\nu \partial_\nu \phi) - \frac{\lambda}{4!} \phi^4 \quad (6)$$

It still retains the $\mathbb{Z}_2$ symmetry $\phi \rightarrow -\phi$, and thus only operators with even powers of ϕ are allowed. For the case where the coupling constant is dimensionless $d = 8$, the by construction the highest power of ϕ that can appear in any term is ϕ^4 , since $[\phi^4] = 4[\phi] = 8$. This leaves only the terms that mix ϕ and its derivative up to that power.

For ϕ^4 term, there can't be any additional derivatives, since every derivative adds a scaling dimension, and ϕ^4 already has the maximal relevant scaling dimension. For ϕ^2 , there are two additional terms: the one with no derivatives, with total scaling dimension $[\phi^2] = 4$, and the one with a single deLambertian (we can only use derivatives in deLambertians $\partial^\mu \partial_\mu$ to preserve Lorentz invariance), which has scaling dimension $[(\partial^\mu \partial_\mu \phi) \phi] = 6$

$$\mathcal{L} = \frac{1}{2} (\partial^\mu \partial_\mu \phi) (\partial^\nu \partial_\nu \phi) - \frac{\lambda}{4!} \phi^4 + g_2 \phi^2 + g_3 (\partial^\mu \partial_\mu \phi) \phi \quad (7)$$

These, of course, match the two terms we removed when setting the mass to zero. In the $m \neq 0$ case, $g_2 = 2m^2$ and $g_3 = m^4$.

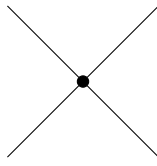
1.3 The scalar propegator in this theory

As the hint implies, we can take the kinetic term, multiply it by 2, then take its inverse in momentum space. This gives:

$$S(k) = \langle \phi(k) \phi(-k) \rangle = \frac{i}{k^4 + 2m^2 k^2 + m^4} \quad (8)$$

1.4 The $2 \rightarrow 2$ scattering amplitude at order λ

The interaction acts similarly to the regular real scalar ϕ^4 interaction, and therefore it has the regular interaction vertex, with the same resultant value:



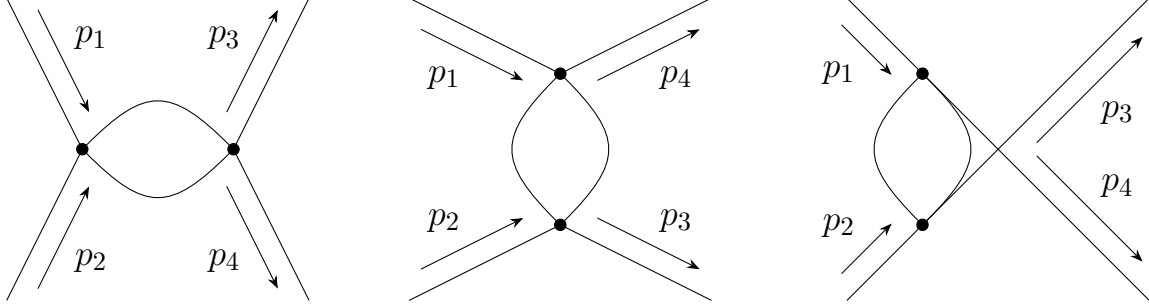
$$= -i\lambda(2\pi)^8 \delta^{(8)} \left(\sum p_i \right) \quad (9)$$

And so:

$$\mathcal{M} = -\lambda \quad (10)$$

1.5 The $2 \rightarrow 2$ scattering amplitude at order λ^2

At the next order, we simply add the three channels s, t, u :



The second order contribution from these three diagrams is:

$$\mathcal{M}_2 = -\lambda^2 [V(s) + v(t) + V(u)] \quad (11)$$

Where:

$$V(p^2) = \frac{1}{2} \int \frac{d^8 q}{(2\pi)^8} \frac{i^2}{(q^4 + 2m^2 q^2 + m^4)((q-p)^4 + 2m^2(q-p)^2 + m^4)} \quad (12)$$

1.6 Calculating

Dimensional Regularization! Woo :)

Identifying the divergences

The degree of the integral is $\frac{d^8 q}{q^8} \sim 1$ which means the integral is logarithmically divergent.

Changing the dimension of the momentum integral $d_0 \rightarrow d$

$$V_d(p^2) = -\frac{1}{2} \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^4 + 2m^2 q^2 + m^4)((q+p)^4 + 2m^2(q-p)^2 + m^4)} \quad (13)$$

Feynman's trick

Using the given formula with

$$\begin{aligned} A &= (q-p)^2 + m^2 & B &= q^2 + m^2 \\ n_1 &= n_2 = 2 \end{aligned} \quad (14)$$

Gives:

$$\begin{aligned} & \frac{1}{(q^4 + 2m^2q^2 + m^4)((q+p)^4 + 2m^2(q-p)^2 + m^4)} = \\ &= \frac{\Gamma(4)}{\Gamma(2)^2} \int_0^1 dx \frac{x^1(1-x)^1}{[x((q+p)^2 + m^2) + (1-x)(q^2 + m^2)]^4} = \\ &= \frac{3}{2} \int_0^1 dx \frac{x(1-x)}{[q^2 + xp^2 + 2xqp + m^2]^4} \end{aligned} \quad (15)$$

This gives:

$$V_d(p^2) = -\frac{3}{4} \int \frac{d^d q}{(2\pi)^d} \int_0^1 dx \frac{x(1-x)}{[q^2 + xp^2 + 2xqp + m^2]^4} \quad (16)$$

Change of variables

Defining $l = q + xp$ and changing the order of integration:

$$V_d(p^2) = -\frac{3}{4} \int_0^1 dx \int \frac{d^d l}{(2\pi)^d} \frac{x(1-x)}{[l^2 + p^2(1-x)x + m^2]^4} \quad (17)$$

Naming the remaining part of the denominator and performing Wick's rotation $l^0 \rightarrow il_E^0$:

$$\Delta = p^2(1-x)x + m^2 \quad (18)$$

Gives us the well known integral:

$$V_d(p^2) = -i \frac{3}{4} \int_0^1 x(1-x) dx \int \frac{d^d l_E}{(2\pi)^d} \frac{1}{[l_E^2 + \Delta]^4} \quad (19)$$

Solving the l_E integral

Using:

$$\frac{d^d l_E}{(2\pi)^d} \frac{1}{(l_E^2 + \Delta)^n} = \frac{1}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(n - \frac{d}{2})}{\Gamma(n)} \Delta^{\frac{d}{2} - n} \quad (20)$$

We get:

$$V_d(p^2) = -i \frac{\Gamma(4 - \frac{d}{2})}{8(4\pi)^{\frac{d}{2}}} \int_0^1 x(1-x) dx \Delta^{\frac{d}{2} - 4} \quad (21)$$

Using dimension $d = 8 - \epsilon$

Set the dimension $d = 8 - \epsilon$:

$$V(p^2) = -i \frac{\Gamma\left(\frac{\epsilon}{2}\right)}{8(4\pi)^4} \int_0^1 x(1-x) dx \left(\frac{\Delta}{4\pi}\right)^{-\frac{\epsilon}{2}} \quad (22)$$

Using the approximation of Γ and of powers of Δ :

$$\Gamma\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon} - \gamma + \mathcal{O}(\epsilon) \quad (23)$$

$$\Delta^{-\frac{\epsilon}{2}} = 1 - \frac{\epsilon}{2} \ln \Delta + \mathcal{O}(\epsilon^2) \quad (24)$$

This gives:

$$V(p^2) = -\frac{i}{8(4\pi)^4} \int_0^1 x(1-x) dx \left(\frac{2}{\epsilon} - \gamma - \ln\left(\frac{\Delta}{4\pi}\right) + \mathcal{O}(\epsilon)\right) \quad (25)$$

Pluggin Δ back in:

$$V(p^2) = -\frac{i}{8(4\pi)^4} \int_0^1 x(1-x) dx \left(\frac{2}{\epsilon} - \gamma - \ln\left(\frac{p^2(1-x)x + m^2}{4\pi}\right) + \mathcal{O}(\epsilon)\right) \quad (26)$$

1.7 Renormalization and the beta function

Defining the physical constant:

$$\lambda_p = \mathcal{M}(s = t = u = M^2) \quad (27)$$

For some energy scale M , we can renormalize the above by simply writing the physical coupling as a function of bare coupling:

$$\lambda_p = \lambda - \lambda^2 \cdot 3V(M^2) + \mathcal{O}(\lambda^3) \quad (28)$$

Flip the expansion:

$$\lambda = \lambda_p + \lambda_p^2 \cdot 3V(M^2) \quad (29)$$

And plug it into the general $2 \rightarrow 2$ scattering amplitude:

$$\begin{aligned} \mathcal{M}_{2 \rightarrow 2} &= \lambda - \lambda^2 [V(s) + V(t) + V(u)] + \mathcal{O}(\lambda^3) = \\ &= \lambda_p - \lambda_p^2 [V(s) + V(t) + V(u) - 3V(M^2)] + \mathcal{O}(\lambda^3) = \\ &= \lambda_p - \frac{\lambda_p^2}{8(4\pi)^4} \int_0^1 x(1-x) dx \left(\ln \frac{x(1-x)s + m^2}{M^2(1-x)x + m^2} + \right. \\ &\quad \left. + \ln \frac{x(1-x)t + m^2}{M^2(1-x)x + m^2} + \ln \frac{x(1-x)u + m^2}{M^2(1-x)x + m^2} \right) + \mathcal{O}(\lambda^3, \epsilon) \end{aligned} \quad (30)$$

Now, to get the beta function, we just need to take the coupling from the flipped expansion, and derive it by M :

$$\begin{aligned}\beta(\lambda) &= M \frac{d\lambda}{dM} = M \lambda^2 \frac{dV(M^2)}{dM} = M \frac{\lambda^2}{8(4\pi)^4} \int_0^1 x(1-x) dx \frac{2M}{x(1-x)M^2 + m^2} = \\ &= \frac{\lambda^2}{1024\pi^4} \int_0^1 \frac{x(1-x)}{x(1-x) + \frac{m^2}{M^2}} dx\end{aligned}\tag{31}$$

For high energy $M^2 \gg m^2$, this becomes:

$$\beta(\lambda) = \frac{\lambda^2}{1024\pi^4}\tag{32}$$

Which is positive, so the coupling constant becomes larger as the energy scale grows, meaning that theory is valid for low energies and is not asymptotically free.

1.8 The RG flow of the massless theory

As we lower the energy cutoff Λ (or raise the energy scale M), some operators will become irrelevant and some will remain. For the massless theory, we chose from the beginwith to only take the relevant operators, those which dimension is smaller then d , so that when the theory scales down, they will grow. So, we reuquire three fine tunings, and the action will look like the one we saw in section 2.